

Magnetic Fields: Introduction

Quick review of electrostatics (why we need “another field”)

1) **Coulomb’s law (force between point charges).** For two point charges q_1 and q_2 located at $\mathbf{r}_1$ and $\mathbf{r}_2$, the force on q_2 due to q_1 is

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{\|\mathbf{r}_2 - \mathbf{r}_1\|^3} (\mathbf{r}_2 - \mathbf{r}_1).$$



Figure 1: Two point charges and Coulomb force (electrostatics).

2) **Electric field $\mathbf{E}$ (superposition).** The electric field due to a collection of point charges is

$$\mathbf{E}(\mathbf{r}) = \sum_i \frac{1}{4\pi\epsilon_0} q_i \frac{\mathbf{r} - \mathbf{r}_i}{\|\mathbf{r} - \mathbf{r}_i\|^3}.$$

For a continuous charge density $\rho(\mathbf{r}')$,

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \rho(\mathbf{r}') \frac{\mathbf{r} - \mathbf{r}'}{\|\mathbf{r} - \mathbf{r}'\|^3} dv'.$$

3) **Potential ϕ and the conservative nature of electrostatics.** The potential due to a charge distribution is

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{\|\mathbf{r} - \mathbf{r}'\|} dv'.$$

In electrostatics,

$$\mathbf{E} = -\nabla\phi, \quad \mathbf{F} = q\mathbf{E}.$$

Also, Gauss’s law:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\epsilon_0}.$$

4) **Key property: path-independence / zero circulation.** For electrostatics,

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0 \quad (\text{closed path})$$

which means the field is conservative and the line integral between two points is path independent.

Electrostatics in one sentence

Electrostatic electric fields are **conservative**:

$$\nabla \times \mathbf{E} = \mathbf{0} \iff \oint \mathbf{E} \cdot d\mathbf{l} = 0 \iff \mathbf{E} = -\nabla\phi.$$

Why do we need to introduce magnetic fields?

Motivation (transition to magnetostatics)

Electrostatics (charges at rest) looks like a “complete” theory. However, when charges **move** (currents) and when fields vary with time, new phenomena appear that cannot be explained using electrostatics alone. This motivates introducing the **magnetic field $\mathbf{B}$** and later the full set of Maxwell’s equations.

Reading note (refer): Feynman Lectures on Physics, Vol. II (Magnetostatics chapter: Chapter 13).

Actually, electrostatics is not a complete theory. Why?

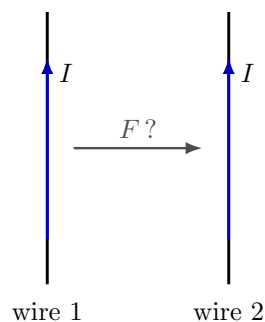


Figure 2: Two long conducting wires carrying current.

Consider two wires driven by current.

What a wire is made of ?

- Wires are made of atoms of a conductor.

- There are fixed positive ion cores (+) forming the lattice.
- There are free electrons ($-e$) that drift to produce current.
- For simplicity, assume ≈ 1 free electron per atom. Let n denote the number density (number of atoms/electrons per m^3 or cm^3).

Variables used (for clarity).

- n : number density of atoms/electrons (cm^{-3}).
- e : elementary charge ($1.6 \times 10^{-19} \text{ C}$).
- $\rho(\mathbf{r})$: volume charge density (C/cm^3).
- ϵ_0 : permittivity of free space.
- $\mathbf{r}$: field (observation) point, $\mathbf{r}'$: source point inside the charge distribution.
- $\mathbf{E}(\mathbf{r})$: electric field at $\mathbf{r}$.

Electrostatics-only viewpoint (why it looks like there should be *no* force). In the lab frame, each small volume of the metal contains

$$\rho_+ = +ne, \quad \rho_- = -ne \implies \rho_{\text{net}} = \rho_+ + \rho_- = 0 \quad (\text{electrically neutral}).$$

Coulomb's law for the electric field due to a total charge density $\rho(\mathbf{r}')$ is

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \iiint \rho(\mathbf{r}') \frac{\mathbf{r} - \mathbf{r}'}{\|\mathbf{r} - \mathbf{r}'\|^3} dv'.$$

So if we (naively) say $\rho(\mathbf{r}') = \rho_{\text{net}}(\mathbf{r}') = 0$ everywhere, then the integral suggests

$$\mathbf{E}(\mathbf{r}) = \mathbf{0} \implies \text{no electric force between the two neutral wires.}$$

But experiments show: the two wires *attract or repel* depending on the current direction (see standard demonstrations; also discussed in many EM texts (e.g., Griffiths)).

So what is missing?

Electrostatics alone (charges-at-rest picture) cannot explain forces between *neutral* current-carrying wires. **Something else must be included.**

Clue: Einstein's special relativity. A key relativistic idea is that when a distribution moves fast, its length along the motion appears contracted (i.e., charges look *more packed per unit length* in some frames).



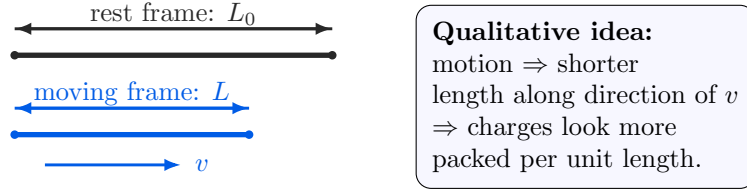


Figure 3: Relativity clue (qualitative): a moving length appears contracted ($L < L_0$), so charges can look “more densely packed” along the motion direction.

We will use this relativity idea to show how a *tiny* charge imbalance can appear in an appropriate frame, leading to an electric-force explanation of wire attraction/repulsion, and motivating the magnetic field concept.

How small is the relativity effect in a real wire? (order-of-magnitude)

Relativity reminder (length contraction + time dilation). If an object moves with speed v (along its length),

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

and (concept of time dilation) a moving clock ticks slower:

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \frac{v^2}{c^2}}}.$$

Numerical estimate using a typical current-carrying wire. Suppose $I = 1$ A passes through a wire of cross-sectional area

$$A = 1 \text{ mm}^2 = 10^{-6} \text{ m}^2.$$

Then the current density is

$$J = \frac{I}{A} = \frac{1}{10^{-6}} = 10^6 \text{ A/m}^2.$$

(Equivalently, since $1 \text{ A/cm}^2 = 10^4 \text{ A/m}^2$, $J = 10^6 \text{ A/m}^2 = 100 \text{ A/cm}^2$.)

Take a typical free-electron number density in a metal as

$$n \sim 10^{22} - 10^{23} \text{ cm}^{-3} \quad (\text{order of atomic number density}).$$

Using

$$J = nev_d$$

(where $e = 1.6 \times 10^{-19} \text{ C}$ and v_d is the drift velocity), and working in cm units:

$$100 \approx (10^{22})(1.6 \times 10^{-19}) v_d \quad \Rightarrow \quad v_d \approx \frac{100}{1.6 \times 10^3} \approx 0.0625 \text{ cm/s}.$$



So

$$v_d \approx 0.0625 \text{ cm/s} \approx 0.6 \text{ mm/s} \quad (\text{very small speed}).$$

Compare to the speed of light.

$$\frac{v_d}{c} \approx \frac{0.0625}{3 \times 10^{10}} \approx 2 \times 10^{-12}.$$

Since $\left(\frac{v_d}{c}\right)^2 \ll 1$, we can use the small- x expansion:

$$\sqrt{1-x} \approx 1 - \frac{x}{2} \quad \text{for } x \ll 1.$$

Putting $x = \left(\frac{v_d}{c}\right)^2$,

$$\sqrt{1 - \left(\frac{v_d}{c}\right)^2} \approx 1 - \frac{1}{2} \left(\frac{v_d}{c}\right)^2 \approx 1 - \frac{1}{2} (2 \times 10^{-12})^2 = 1 - 2 \times 10^{-24}.$$

So the length change is extremely tiny. Hence

$$L = L_0 \sqrt{1 - \frac{v_d^2}{c^2}} \approx L_0 (1 - 2 \times 10^{-24}) = L_0 - 2 \times 10^{-24} L_0.$$

The contraction is tiny, but the idea still matters for understanding current-carrying wires.

What relativity changes in a current-carrying wire

- Here, the moving particles are **electrons** (drift with v_d).
- Due to relativity, the *spacing along the direction of motion* appears length-contracted in an appropriate frame, so the *number of charges per unit length* (linear density) appears slightly larger.
- The positive ion cores (atoms) are taken as immovable in the lab frame (lattice fixed), so their density changes differently depending on the chosen reference frame.

So what is the new situation with the relativity effect? Think in terms of charge density along the wire. In the lab frame, we usually take the wire as neutral:

$$\lambda_+ + \lambda_- = 0,$$

where λ_+ and λ_- are the $+$ and $-$ **linear** charge densities (C/m).

Now, in a frame where electrons are closer to rest (“electron viewpoint”), the ion lattice is moving. Because moving lengths are slightly contracted, the moving lattice appears *slightly more packed per unit length*. Therefore, the $+$ density seen in that frame is *slightly larger* than what was balanced in the lab frame, which produces a **tiny net charge imbalance** (very small, but not exactly zero).



In summary:

So in both wires carrying current, a *tiny* fraction of net charge per unit length is effectively built up ($\lambda_{\text{net}} \neq 0$ in the appropriate moving viewpoint).

Why can this create a force between the two wires? (Coulomb's law idea) Draw the two wires and indicate that, in the moving viewpoint, one wire can look slightly charged, hence it can attract the electrons of the other wire via Coulomb's law.

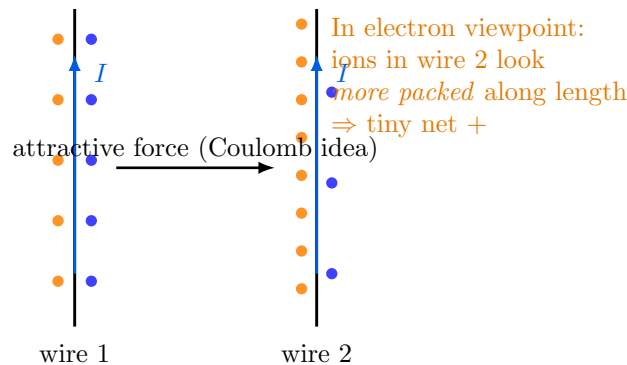


Figure 4: In an appropriate moving viewpoint, one wire can appear slightly charged $\Rightarrow$ Coulomb attraction/repulsion is possible.

Parallel-car analogy We also need a “point of view” statement:

Parallel-car analogy (exact role in our argument)

Assume two cars move in parallel with the same speed v . If you sit in one car, you feel *you are at rest* and the *road moves backward*, not that “your car is moving”. Similarly, for electrons in wire 1, it is natural to describe other charges using the electron’s viewpoint.

Which reference frame is moving with respect to what?

We will use three viewpoints Let the electron drift speed be v_d (very small), along $+\hat{z}$.

Frames we will refer to:

- **Frame S (Lab frame):** ion lattice of both wires is at rest; electrons drift with speed v_d . Each wire is electrically neutral: $\lambda_+ + \lambda_- = 0$.
- **Frame S'_1 (electron viewpoint of wire 1):** move with the electrons of wire 1. Then electrons in wire 1 are (nearly) at rest, while the ion lattice of wire 1 moves backward.
- **Frame S'_2 (electron viewpoint of wire 2):** similarly, move with electrons of wire 2. (Used to argue symmetry: if wire 1 feels a force due to wire 2, wire 2 feels an equal and opposite force.)

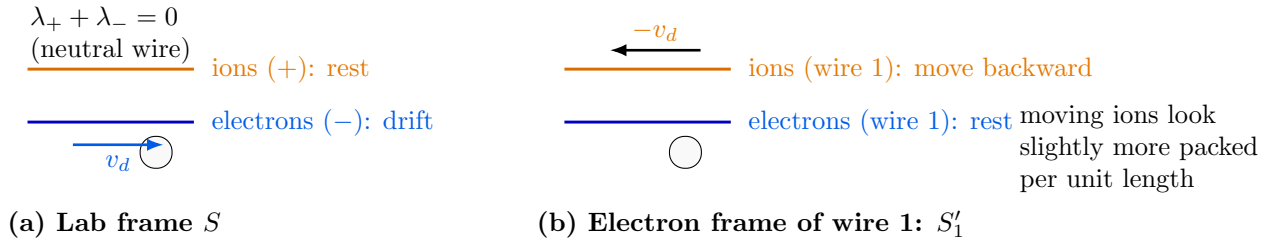


Figure 5: Same physical wire described in two frames. In the electron viewpoint, the ion lattice is moving, so its *linear density* appears slightly different (length-contraction idea).

Now apply this viewpoint idea to *two* wires.

1. Why there is no “net force” issue inside the same wire.

Inside a single wire, electrons and ions do exert forces on each other, but these are *internal* forces. The overall wire remains in mechanical equilibrium (and any internal stresses are supported by the lattice). The important point for us is the **external interaction between wire 1 and wire 2**.

2. Why the other wire can look slightly charged in the electron viewpoint.

In S'_1 (electron frame of wire 1):

- wire 1 electrons are at rest; wire 1 ions move backward $\Rightarrow$ their linear density shifts slightly.
- For **parallel currents**, electrons in wire 2 drift in the *same* direction as in wire 1, so in S'_1 they are also *almost at rest*, while wire 2 ions still move backward.
- That makes the + line in wire 2 appear *slightly more packed* than the – line (in this viewpoint), so wire 2 has a **tiny net** + linear charge density in S'_1 .

Therefore, in S'_1 there is a small $\mathbf{E}$ field due to wire 2, and electrons in wire 1 feel a Coulomb force *toward* wire 2 $\Rightarrow$ **attraction**.

3. Why repulsion happens when currents are opposite.

If wire 2 current is reversed, then in S'_1 the electrons of wire 2 are *not* at rest. The “packing”

comparison flips in sign, giving a tiny net charge of opposite sign in that viewpoint, so the Coulomb force reverses $\Rightarrow$ **repulsion**.

Important remark. The imbalance is extremely small (it scales with $(v_d/c)^2$), but its *direction* is enough to explain:

parallel currents attract, anti-parallel currents repel.

Conclusion:

If we include special relativity, Coulomb's law can explain forces between neutral current-carrying wires.

However, doing everything with changing reference frames becomes cumbersome. So we introduce a new concept—the **magnetic field**—to describe current-related forces directly and cleanly.

We have known magnets for a long time

Compass and the Earth's magnetic field. A freely suspended bar magnet (compass needle) aligns with the Earth's magnetic field. The end marked “N” (north-seeking pole) points approximately toward *geographic north*, because geographic north lies near the Earth's *magnetic south* pole. Equivalently, the needle aligns along the local $\mathbf{B}_{\text{Earth}}$ direction, which (outside the Earth) points from the Earth's magnetic N pole toward its magnetic S pole.

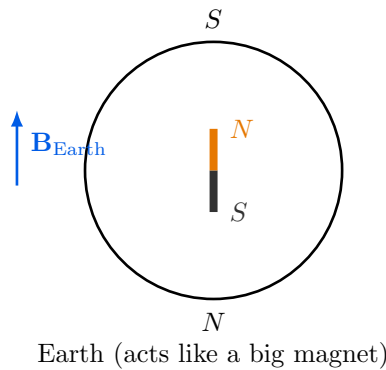


Figure 6: Correct alignment: with Earth's magnetic S at the top, $\mathbf{B}_{\text{Earth}}$ points upward (outside field $N \rightarrow S$), so the needle's N end points upward.

Two magnets: attraction/repulsion and alignment. When two bar magnets are brought close,

$N-S$ attract, $N-N$ repel, $S-S$ repel.

They also tend to rotate so that they align in a low-energy configuration (often N facing S).



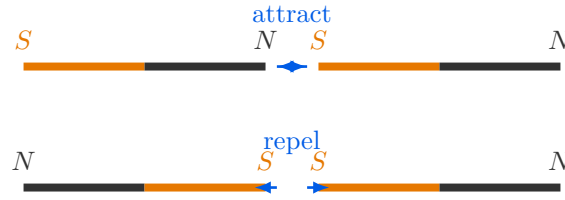


Figure 7: Two bar magnets: unlike poles attract, like poles repel; magnets also rotate to align.

How does magnetic force fall with distance?

Experimentally, we observe that the magnetic force between two magnets *decreases rapidly* with separation, and in a simple (Coulomb-like) picture we often describe it as approximately

$$F \propto \frac{1}{r^2}.$$

Note: This $1/r^2$ is an *approximate description* (useful for intuition). The exact distance dependence depends on the magnet shape and the separation regime; we will treat those details later in the course.

Iron filings experiment: field-line pattern. A classic school experiment: sprinkle iron filings around a bar magnet. The filings align and reveal the *pattern* of the magnetic field around the magnet.

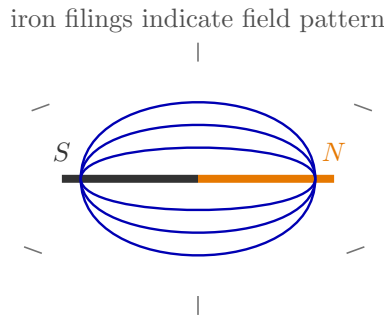
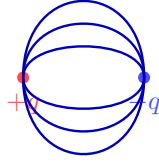


Figure 8: Iron filings align along the magnetic field pattern around a bar magnet (schematic).

Connection to what we already know: electric dipole vs magnetic dipole (pattern similarity).

We already studied electric dipoles in electrostatics: two equal and opposite charges $\pm q$ separated by a small distance. Their electric field line pattern resembles the field pattern around a bar magnet.

electric dipole



bar magnet (magnetic dipole)

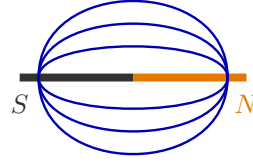


Figure 9: Field-line *patterns* look similar for an electric dipole and a bar magnet (schematic).

Important clarification (to avoid a common misconception)

The *patterns* are similar, but the **physics is not identical**: electric field lines begin/end on charges ($\pm q$), whereas magnetic field lines form **closed loops** (no isolated magnetic charge has been observed in standard electromagnetism).

From magnets to currents in wires (historical motivation)

1) A freely hanging magnet aligns with Earth's field. A freely suspended bar magnet (compass needle) aligns with the Earth's magnetic field. The end marked *N* (north-seeking pole) points approximately toward *geographic north* (geographic north lies near the Earth's *magnetic south* pole).

freely hanging magnet

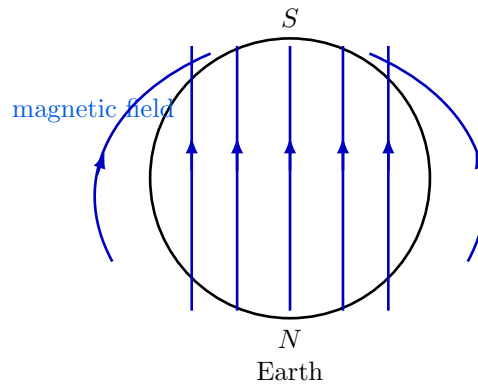
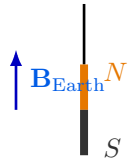


Figure 10: A freely hanging magnet aligns with the Earth's magnetic field. Outside the Earth, field lines go from the Earth's magnetic *N* pole to its magnetic *S* pole, so the needle's *N*-end points toward the Earth's magnetic *S* (near geographic north).

2) Bring a current-carrying wire near the magnet. Now consider bringing a *current-carrying wire* near the magnet. Instead of aligning only with the Earth's field, the magnet starts aligning with the field produced by the current in the wire.



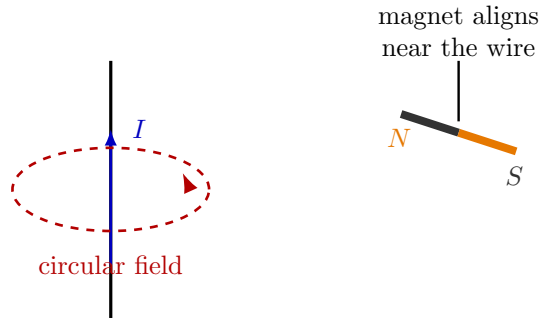


Figure 11: Observation: a current-carrying wire produces a circular magnetic field around it, so a nearby magnet tends to align accordingly.

Key statement:

A wire carrying current produces a **circular magnetic field** around it.

Biot–Savart law (empirical law from current–magnet experiments)

After collecting data from many current–magnet experiments, the mathematical rule they arrived at can be stated as follows:

Consider a *small current element* $I d\ell$ located at the *source point* $\mathbf{r}'$. We want the small magnetic-field contribution $d\mathbf{B}$ at the *field point* $\mathbf{r}$.

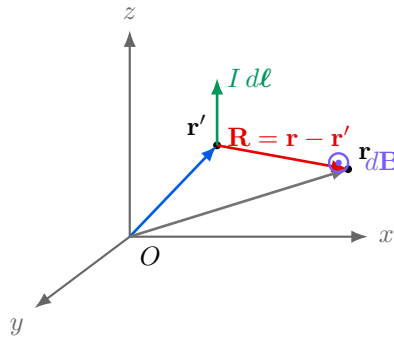


Figure 12: Biot–Savart geometry (less cluttered): $I d\ell$ taken along $+\hat{\mathbf{z}}$ at $\mathbf{r}'$, field point $\mathbf{r}$, and $\mathbf{R} = \mathbf{r} - \mathbf{r}'$.

The *magnitude* is observed to depend on:

$$dB \propto \frac{I d\ell \sin \theta}{R^2}, \quad R = |\mathbf{R}|,$$

and the *direction* is perpendicular to the plane containing $d\ell$ and $\mathbf{R}$ (right-hand rule).

Biot–Savart law (vector form)

$$d\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{I d\boldsymbol{\ell} \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}$$

Variables (as used above).

- $\mathbf{r}$: field point where we want $\mathbf{B}$.
- $\mathbf{r}'$: source point (location of the small current element).
- $\mathbf{R} = \mathbf{r} - \mathbf{r}'$, $R = |\mathbf{R}|$: vector (and distance) from source to field point.
- I : current in the wire.
- $d\boldsymbol{\ell}$: directed infinitesimal length element along the wire (direction of current).
- μ_0 : permeability of free space (constant).
- θ : angle between $d\boldsymbol{\ell}$ and $\mathbf{R}$.

Magnetic field of an (infinitely) long straight wire (Biot–Savart)

Consider a long straight wire carrying current I along $+\hat{\mathbf{z}}$. We want $\mathbf{B}$ at a point P located a perpendicular distance r from the wire (on the x -axis for convenience).

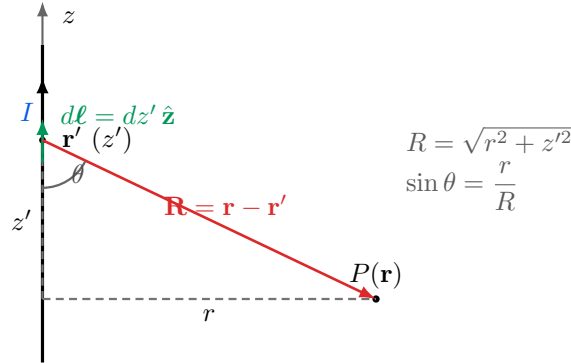


Figure 13: Geometry for a long straight wire: a small element $I d\boldsymbol{\ell}$ at $\mathbf{r}'$ produces $d\mathbf{B}$ at $P(\mathbf{r})$. Here θ is the (acute) angle between $\mathbf{R}$ and the wire direction.

Step 1: Write Biot–Savart for a current element.

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\boldsymbol{\ell} \times \mathbf{R}}{R^3}.$$

Here $d\boldsymbol{\ell} = dz' \hat{\mathbf{z}}$ and $R = \sqrt{r^2 + z'^2}$. The magnitude of the cross product is

$$|d\boldsymbol{\ell} \times \mathbf{R}| = dz' R \sin \theta.$$



Step 2: Replace $\sin \theta$ using the triangle. From the right triangle,

$$\sin \theta = \frac{r}{R} = \frac{r}{\sqrt{r^2 + z'^2}}.$$

So the magnitude becomes

$$dB = \frac{\mu_0}{4\pi} \frac{I dz' R \sin \theta}{R^3} = \frac{\mu_0}{4\pi} \frac{I dz' r}{(r^2 + z'^2)^{3/2}}.$$

The direction is the same for all elements (right-hand rule), so

$$d\mathbf{B} = \hat{\phi} \frac{\mu_0}{4\pi} \frac{I r dz'}{(r^2 + z'^2)^{3/2}}.$$

Step 3: Integrate along the whole wire (infinite length).

$$\mathbf{B}(r) = \hat{\phi} \frac{\mu_0 I r}{4\pi} \int_{-\infty}^{+\infty} \frac{dz'}{(r^2 + z'^2)^{3/2}}.$$

Compute the integral (standard result):

$$\int \frac{dz'}{(r^2 + z'^2)^{3/2}} = \frac{z'}{r^2 \sqrt{r^2 + z'^2}}.$$

Therefore,

$$\int_{-\infty}^{+\infty} \frac{dz'}{(r^2 + z'^2)^{3/2}} = \left[\frac{z'}{r^2 \sqrt{r^2 + z'^2}} \right]_{-\infty}^{+\infty} = \frac{1}{r^2} (1 - (-1)) = \frac{2}{r^2}.$$

Final result.

$$\mathbf{B}(r) = \hat{\phi} \frac{\mu_0 I}{2\pi r}$$

Remark:

For a long straight wire, the magnetic field lines form **circles** around the wire, and the magnitude falls as $B \propto 1/r$.

Force on a moving charge in a magnetic field (and the v^2/c^2 connection)

1) Start with the observed force law (Lorentz magnetic force). A charge q moving with velocity $\mathbf{v}$ in a magnetic field $\mathbf{B}$ experiences a force

$$\mathbf{F}_B = q (\mathbf{v} \times \mathbf{B})$$

(Direction is perpendicular to both $\mathbf{v}$ and $\mathbf{B}$, right-hand rule.)

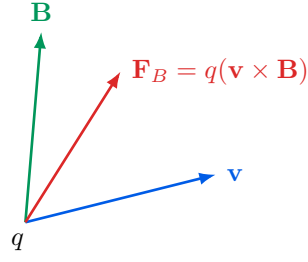


Figure 14: Magnetic force direction: $\mathbf{F}_B \perp \mathbf{v}$ and $\mathbf{F}_B \perp \mathbf{B}$.

2) From many charges to current density Suppose there are n mobile charges per unit volume (number density), each with charge q and average drift velocity $\mathbf{v}_d$. In a small volume element V ,

$$\mathbf{F}_{\text{on charges in } V} = (nV) q (\mathbf{v}_d \times \mathbf{B}).$$

Define the *current density* (A/m²):

$$\mathbf{J} = nq\mathbf{v}_d$$

Then the magnetic force on the charges in volume V can be written as

$$\mathbf{F} = (\mathbf{J} \times \mathbf{B}) V$$

(Equivalently, the force *density* is $\mathbf{f} = \mathbf{J} \times \mathbf{B}$.)

3) Where does the v^2/c^2 idea come in? (Link to your relativity discussion). From the earlier “two current-carrying wires” discussion:

- In the lab frame, each wire is *overall neutral* (ions + and electrons – nearly cancel).
- Electrons drift with a small speed v_d , but they are *moving charges*.
- Special relativity says lengths along the direction of motion are slightly changed. That means *charge densities* can look slightly different in different frames.

The key point is: the relativistic correction is small and scales like

$$(\text{tiny imbalance in density}) \propto \frac{v_d^2}{c^2}, \quad \text{when } v_d \ll c.$$

So the *electric* interaction between wires due to that tiny density imbalance also scales like

$$F_{\text{between wires}} \propto \frac{v_d^2}{c^2} \times (\text{Coulomb-type factor}).$$

This is exactly the qualitative statement: $F \propto (v^2/c^2) \frac{1}{d^2}$ (up to geometry constants).



4) **The main idea (what I want students to remember).** Magnetism appears when charges are moving. At everyday drift speeds ($v_d \ll c$), the effect is small in a *relativistic* sense, but it is still measurable and very important.

Remark (connection to the earlier v^2/c^2 discussion)

The “extra” force you see between current-carrying conductors can be understood as a *relativistic correction* to electrostatics: a tiny change in charge density of order $\sim v_d^2/c^2$ can produce a measurable interaction. Instead of doing relativity every time, we introduce a new field $\mathbf{B}$ and use

$$\boxed{\mathbf{F}_B = q(\mathbf{v} \times \mathbf{B})} \quad \text{and} \quad \boxed{\mathbf{f} = \mathbf{J} \times \mathbf{B}}.$$